



MCT231 BASICS OF MECHATRONICS

Final Project Submission Report

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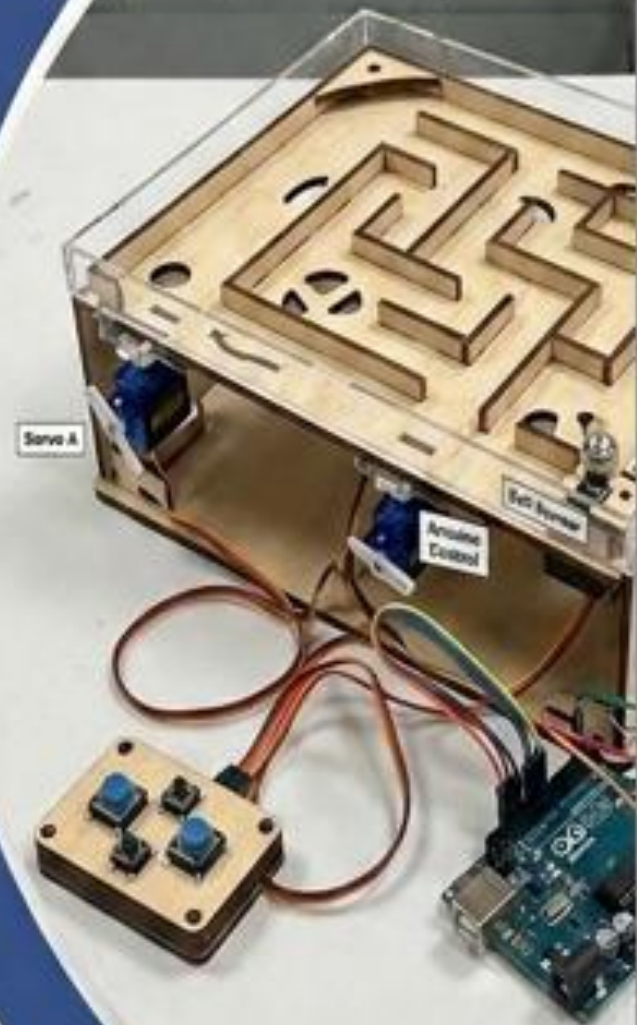


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Introduction:

Designing and developing a mechatronic system requires the effective integration of mechanical design, electronics, and software control into a unified and functional solution. This Major Task project reflects the application of core engineering principles and practical skills acquired throughout the course, with a focus on hands-on development, system integration, and iterative prototyping.

The main objective of the project is to conceptualize and implement a complete system that performs a defined function reliably and efficiently. The development process follows a structured design methodology, beginning with problem identification and requirement analysis, followed by concept generation, detailed design, prototyping, and final implementation. The system incorporates key subsystems such as actuation mechanisms, embedded control units, and sensing components working together to achieve the desired performance.

The project brings together multiple engineering domains, including microcontroller programming, electronic circuit design and PCB development, mechanical modeling and assembly, and system simulation. User interaction and communication interfaces are also considered to enhance system control, monitoring, and feedback.

All stages of the design and development process are presented, including the reasoning behind major decisions, testing and validation procedures, and performance evaluation. Challenges encountered during implementation are discussed along with the solutions applied, ensuring that the final system meets functional requirements as well as relevant engineering and safety standards.

Work Break Structure (WBS):

1	Project Initiation				
	1.1	Project Planning			
		1.1.1	Gantt Chart Creation		
		1.1.2	Defining deliverables		
2	Mechanical Design				
	2.1	Designing the mechanisms			
		2.1.1	Sorting mechanism		
		2.1.2	Vertical lifting mechanism		
		2.1.3	Projectile mechanism		
			2.1.3.1	Projectile calculations	
	2.2	Actuator Sizing			
		2.2.1	Torque/Force calculations		
		2.2.2	Motor/Servo selection		
	2.3	Fully integrated CAD Design			
3	Coding				
	3.1	Mechanical logic coding			
	3.2	Electrical logic coding			
4	Electrical and Control Design				
	4.1	Microcontroller integration			
	4.2	Sensor integration			
	4.3	User interface design			
		5.3.1	LCD display		
		5.3.2	Interactive mobile app		
	4.4	PCB design			
	4.5	Wiring and connections			
5	Prototyping and Testing				
	5.1	Prototype assembly			
	5.2	Testing			
	5.3	Iterations			

Gantt Chart

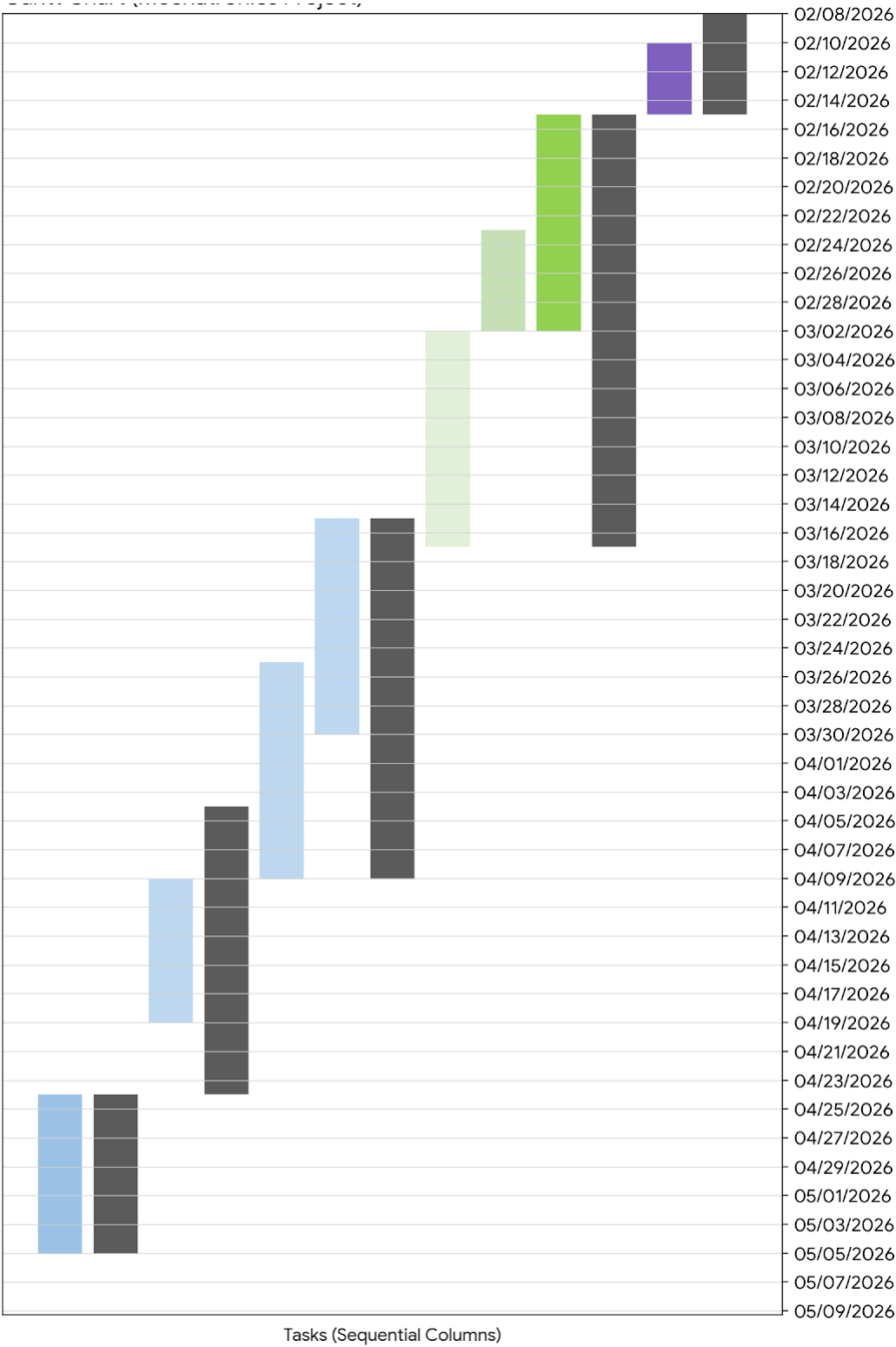


Figure 1 Gantt Chart

Project Design Specification (PDS)

1. Introduction

“Help. Save Buttercup!” is a machine that automatically lifts, sorts, and shoots a ball at targets chosen by the user. It works in a continuous loop, using sensors and motors to keep the ball moving without needing someone to manually reset it.

2. Basics Operation

First, a turnstile drops one ball at a time into a color sorter. A servo motor then guides the ball toward one of two distinct paths based on its color that lead to a ball shooter. Next, the user gives a command to shoot the ball at one of two targets. After the ball hits a target, it rolls to an arm lifter. A sensor sees the ball and turns on a motor to lift it back to the top turnstile. The cycle then starts all over again.

3. Scope

- **Intentions & Audience:** This machine is a sophomore-level project for the Basics of Mechatronics course at Ain Shams University. It is built to show how sensors, coding, and moving parts work together in a continuous loop. The Audience is from 8-10 years of age.
- **Restrictions:** The design is limited to using only two Arduinos and one power supply. It also needs to be finished before the semester ends.

4. Functional Requirements

a) Construction

Key Consideration	Requirements	Issues/Challenges
Materials and process	• 3D printed plastic (PLA) for the ball turnstile, sorter gates, and custom mounts. • Acrylic or wood (MDF) for the main frame and tracks. • DC motor for the shooter mechanism. • Processes: 3D printing, laser cutting, and manual assembly.	• 3D printed parts must have precise tolerances, so the ball doesn't get stuck. • The DC motor must be strong enough to shoot the ball, but not so strong that it breaks the wood mounts.
Dimensions and weight	• Height: Max 60 cm • Width: Max 1m • Length: Max 1m • Weight: Max 10kg	• The entire machine must fit inside the 1m by 1m square footprint.
Quantity	Initial production of 1 working prototype unit.	Since you are only building one machine, if a custom part breaks during testing, the team has to stop working to make a new one.
Lifespan/ Reliability	The project is expected to work correctly for 1 year (to survive grading and testing). • It must run smoothly in a continuous loop without the ball jamming.	• The single power supply and the two Arduinos might get hot if you run the loop for too long without a break. • Wires can easily vibrate loose while the motors are running.
Maintenance	• Check all wire connections before turning the power on. • Wipe dust off the color and IR sensors so they read correctly.	• If you pack all the parts too tightly inside the 1m by 1m space, it will be hard to reach your hands inside to fix a broken wire or clean a sensor.
Power consumption	Total required ampere by the system = 4080mA Power consumption for the entire system= 24.04W A 12V 3A power supply will be used to power the machine	The entire system must be power with a single power supply

b) Experience

Key Consideration	Requirements	Issues/Challenges
Aesthetics	• The machine should have a clean, organized look with neatly tied wiring (good cable management). • Different sections (Sorter, Shooter, Lifter) should be color-coded to match the theme	• Keeping all the wires from the two Arduinos and the shared power supply tidy without making them too hard to reach for repairs.
Ergonomics	• The controls (buttons or mobile interface) used to select a target and shoot must be clearly labeled and easy to press. • The LCD screen displaying the score from the IR sensors must be positioned at an easy-to-read angle.	• Making sure there is no noticeable delay between the user pressing the "shoot" button and the motor actually firing the ball. • Placing the user controls in a safe spot away from the moving turnstile.

c) Safety

Key Consideration	Requirements	Issues/Challenges
Safety considerations	• Moving parts like the lifter's arm, the turnstile, and the shooter's can pinch fingers . • The shooter mechanism could accidentally fire the ball over the 60 cm height limit or outside the machine. • The shared power supply and DC motors could overheat during continuous loops.	• Making sure users and teammates keep their hands out of the 1m by 1m area while the system is powered on.
Safety standards	• All wiring between the two Arduinos, sensors, and motors must be fully covered (insulated) to prevent short circuits . • The system must run entirely on low-voltage DC power.	• Designing covers for the moving parts that protect the user, but still allow you to see the ball moving through the sorter and lifter.
Testing	• Run the power supply and motors for a continuous 10-minute loop to check for overheating before the final presentation. • Test-fire the shooting mechanism at least 20 times to ensure the ball always hits the IR targets and never flies out of bounds.	• Safely testing the maximum force of the shooter's DC motor without breaking the 3D-printed parts or the targets.

d) Transport

Key Consideration	Requirements	Issues/Challenges
Packaging	• The machine must be built on a solid, rigid base (like thick wood) so the frame does not bend when you pick it up.	• A 1m by 1m board is quite wide, making it tricky to carry through standard doorways without bumping the edges.

	• Sensitive parts like the Arduinos, LCD screen, and sensors should be taped down or covered so they don't get crushed.	• The 60 cm height means the lifter or shooter might catch on things if not careful.
Shipping	• The project must fit into the back seat or trunk of a standard car or SUV to be transported to the university lab. • It needs to be carried flat to keep the balls from falling out of the tracks.	• Bumps during the car ride could easily shake the Arduino wires loose or mess up the exact 110° alignment of the turnstile servo.

e) Environmental

Key Consideration	Requirements	Issues/Challenges
Recycling	• At least 80% of the electronic parts (the 2 Arduinos, servos, DC motors, LCD, and sensors) must be easily removable to be reused in future projects. • The main 1m x 1m base should be kept intact for another team or project if possible.	• If you use too much hot glue or permanent adhesive to attach the motors and sensors, you might break them when trying to take the machine apart.
Disposal	• Any broken wires, burnt-out components, or damaged batteries must be thrown away in designated electronic waste (e-waste) bins. • Broken 3D-printed plastic parts can be thrown in the standard trash.	• Making sure the team actually takes the time to unscrew and save the parts after grading, rather than just throwing the whole 1m by 1m board in the dumpster.

Fully Integrated System

1. CAD Design

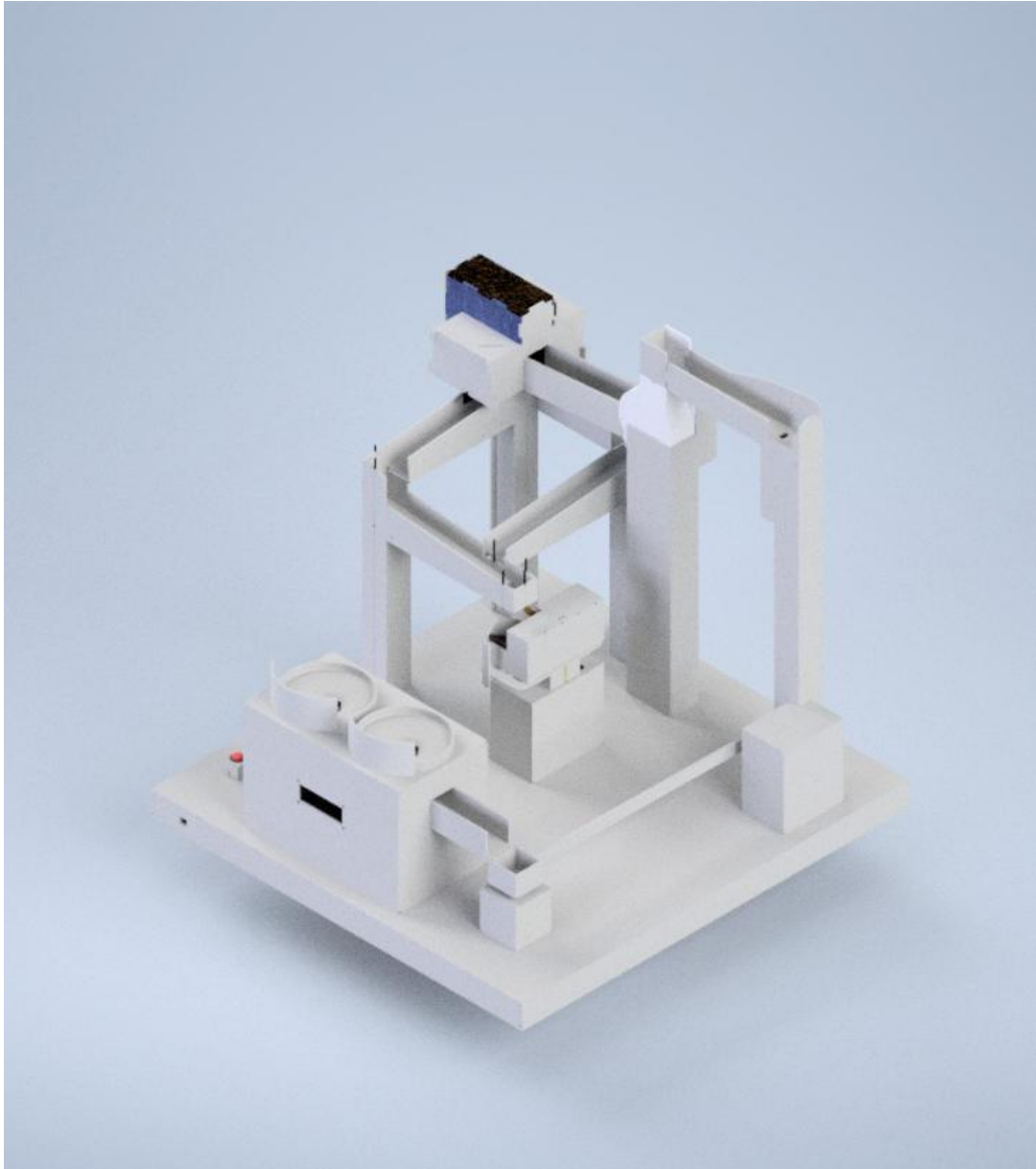


Figure 2 Fully Integrated CAD design

2. Wiring Diagram

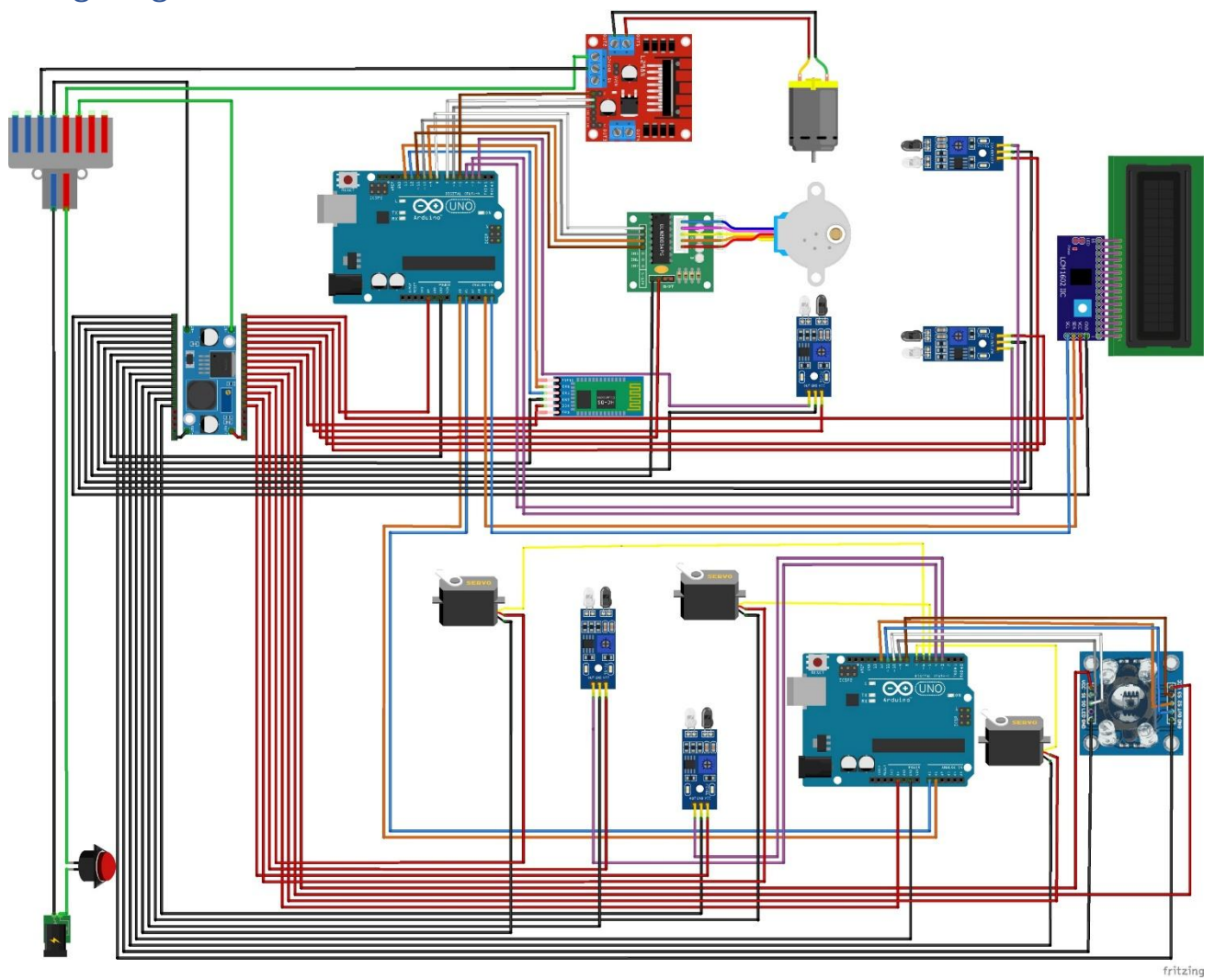
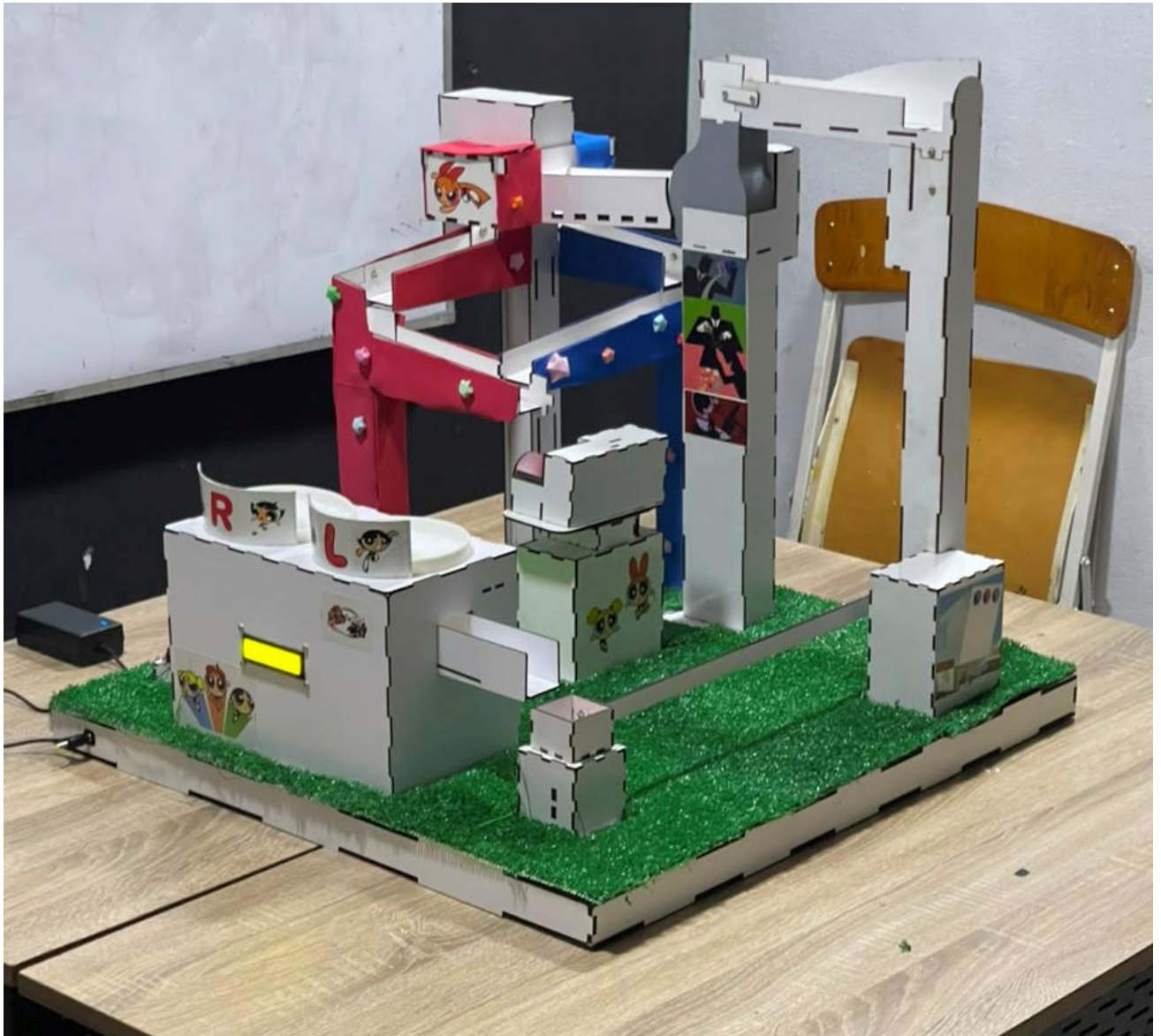


Figure 3 Fully integrated Wiring Diagram

The flowchart illustrates the power-on sequence for a power plant, starting with 'START (Power ON)'. It branches into two main paths: 'Activate 1 (Main Controller)' and 'Activate 2 (Shooter Controller)'. The 'Activate 1' path involves checking for 'Error message 1', and if none, it proceeds to 'Activate Ball Launcher', 'Ball Launcher', 'Shooter', and 'Lifter'. The 'Activate 2' path involves checking for 'Error message 2', and if none, it proceeds to 'Shooter'. Both paths converge at 'Check power'. A 'Safety Consideration' section details the procedure if power is cut while the system is running, involving a series of checks and actions to safely shut down the system, including stopping the ball launcher, shooter, and lifter, and checking for 'Error message 3'.

5. Hardware Implementation

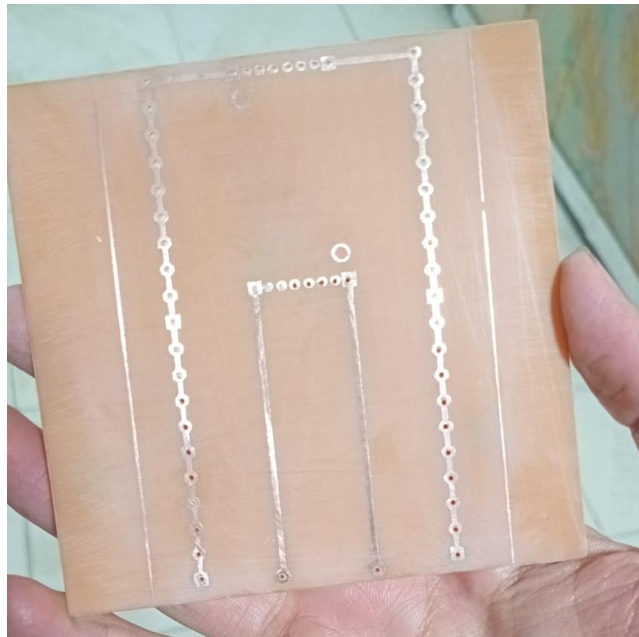
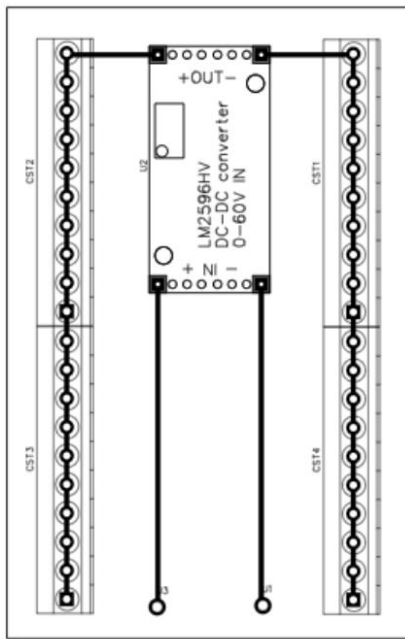


6. Arduino Code

The entire system code can be found on the GitHub repository attached below:

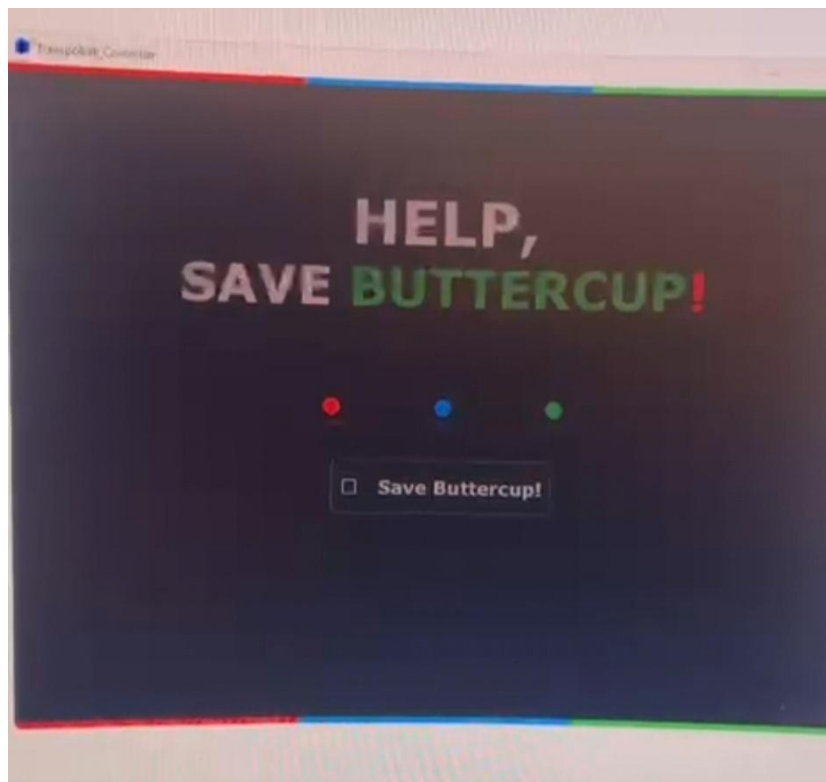


7. PCB



The above PCB was used to distribute a voltage of 5V on various components using a buck converter. The converter steps down the voltage from 12V received by the supply to 5v that is more manageable for most electronic devices. Ewe attached terminal blocks in the pin holes on the PCB for a more professional wire management

8. Processing I/O user interface



We made a user interface that controlling the shooting mechanism via Bluetooth to add an interactive feature to the game.

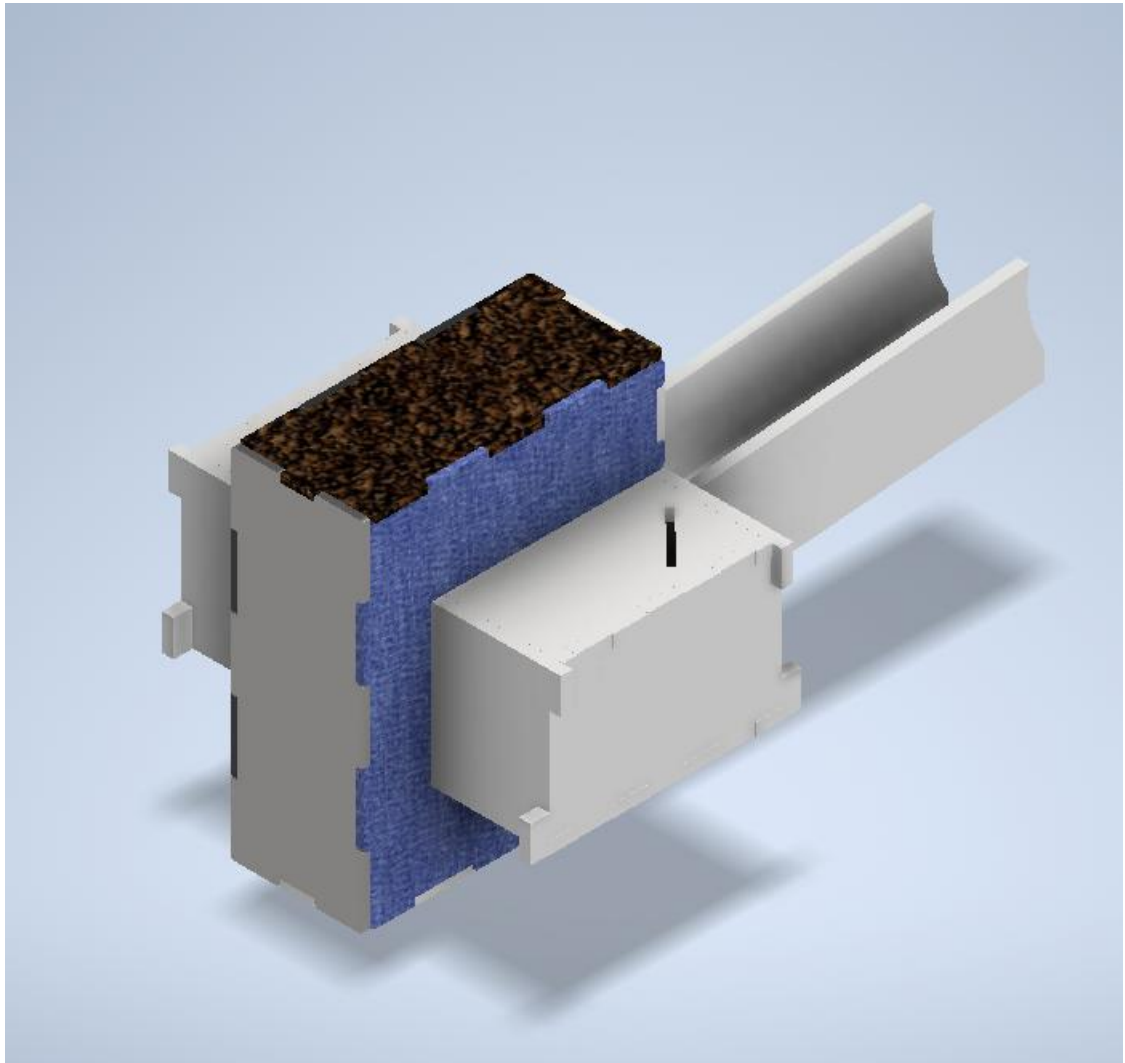
9. LCD



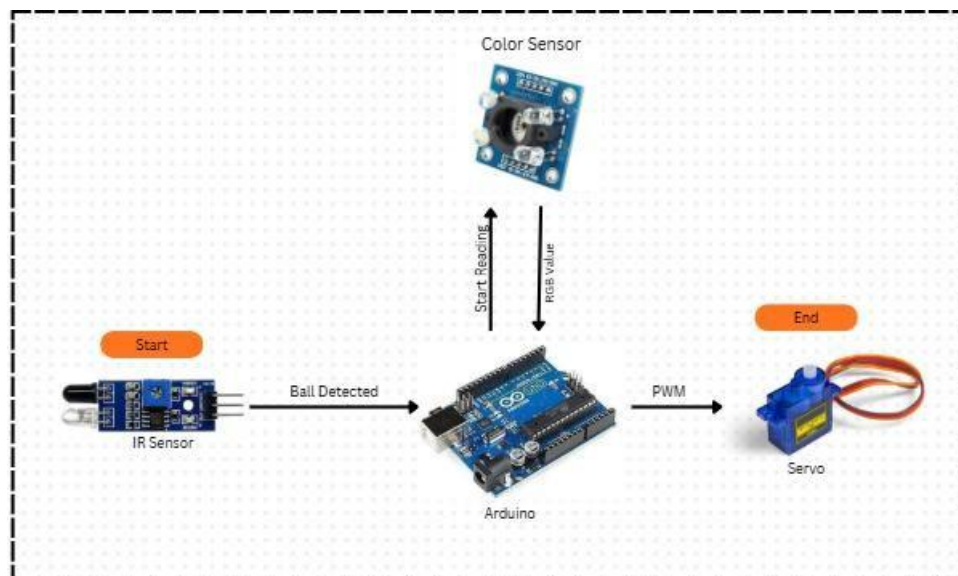
The LCD on the shooter box serves as an HP meter for the game, where you win by dealing damage through the I/O interactive feature.

Sorting Mechanism

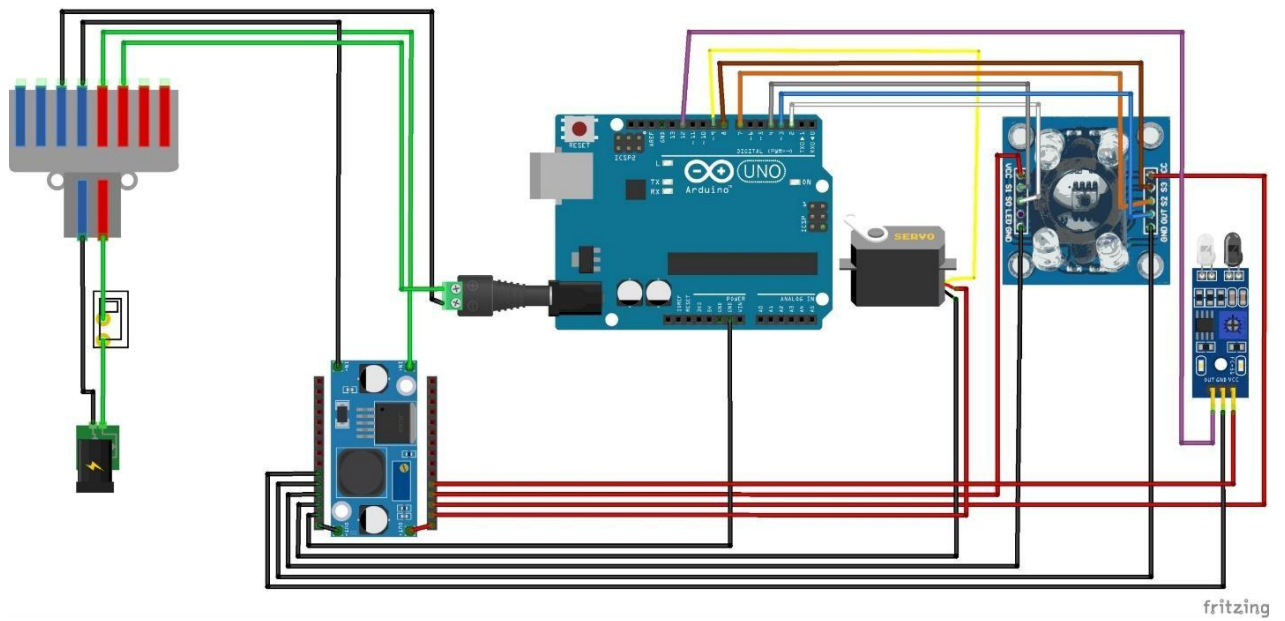
1. CAD design



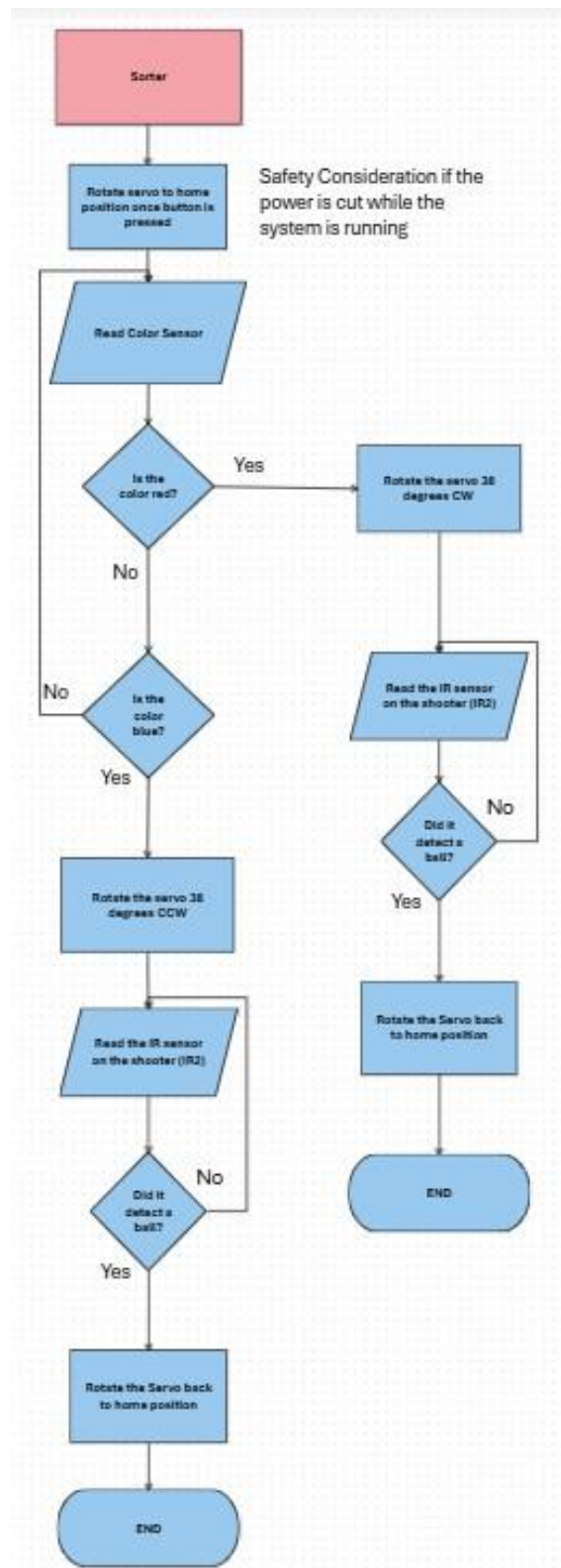
2. Block Diagram



3. Wiring Diagram



4. Flowchart



5. Actuator sizing

$$m_{\text{chuck}} = 0.015 \text{ kg} \quad \text{Sorter}$$

$$m_{\text{ball}} = 0.006 \text{ kg}$$

$$r_{\text{chuck}} = \sqrt{(-14.703)^2 + (-32142)^2} = 35.35 \approx 0.03535 \text{ m}$$

$$r_{\text{ball}} = 0.055 \text{ m}$$

$$J_{\text{chuck}} = 0.015 (0.03535)^2 = 1.874 \times 10^{-5} \text{ kg} \cdot \text{m}^2$$

$$J_{\text{ball}} = 0.006 (0.055)^2 = 1.815 \times 10^{-5} \text{ kg} \cdot \text{m}^2$$

$$J_{\text{total}} = 3.689 \times 10^{-5} \text{ kg} \cdot \text{m}^2$$

$$45^\circ = 0.785 \text{ rad}, \text{ time} = 0.25$$

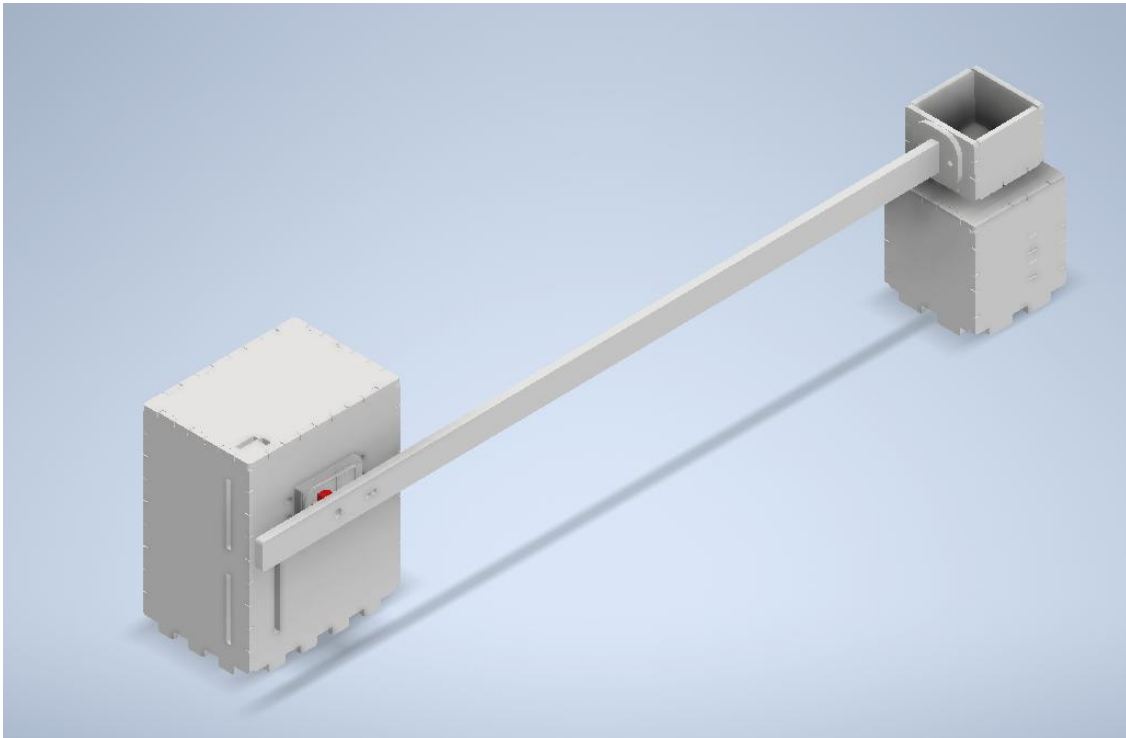
$$\alpha = \frac{40}{\text{r}^2} = \frac{4(0.785)}{(0.25)^2} = 50.24 \text{ rad/s}^2$$

$$T_{\text{req}} = J_{\text{total}} \cdot \alpha = 3.689 \times 10^{-5} (50.24) = 0.00185 \text{ N} \cdot \text{m}$$

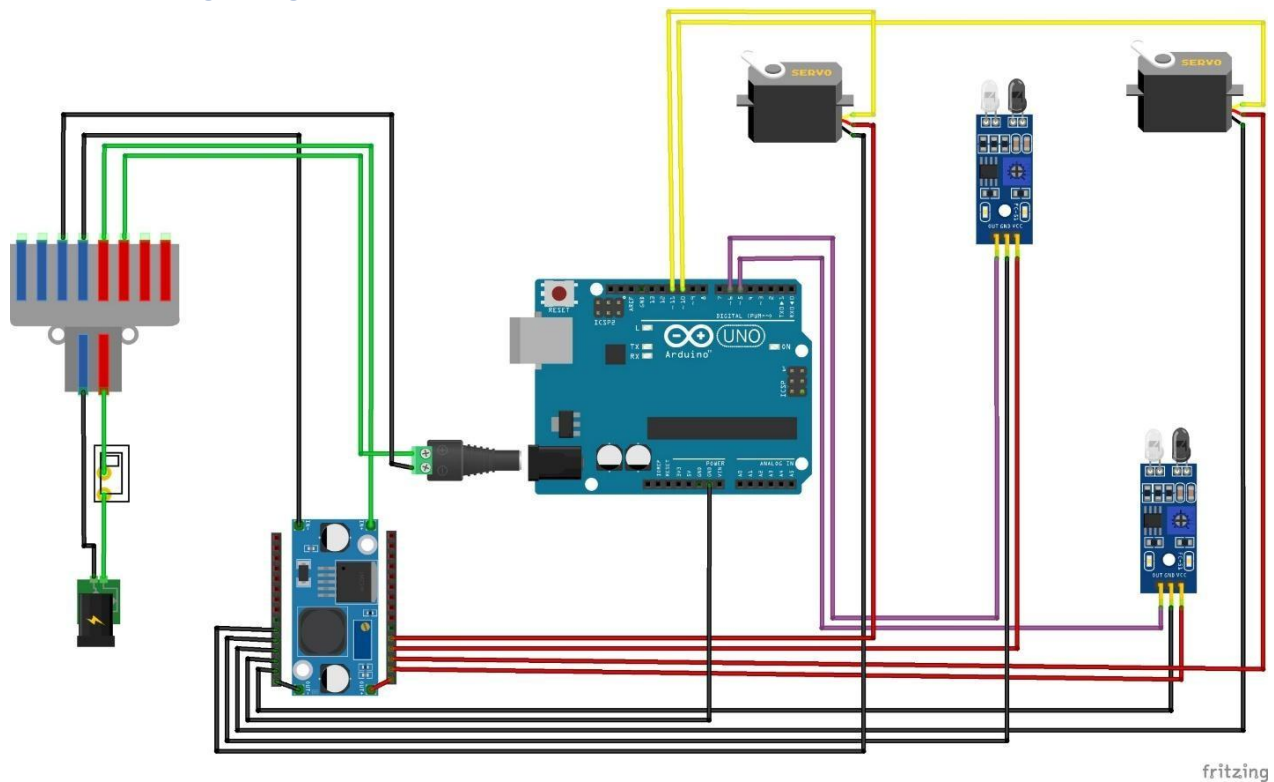
$$\text{Safety Factor} = 0.00185 \times 1.5 = 0.002775 \text{ N} \cdot \text{m}$$

Lifting Mechanism

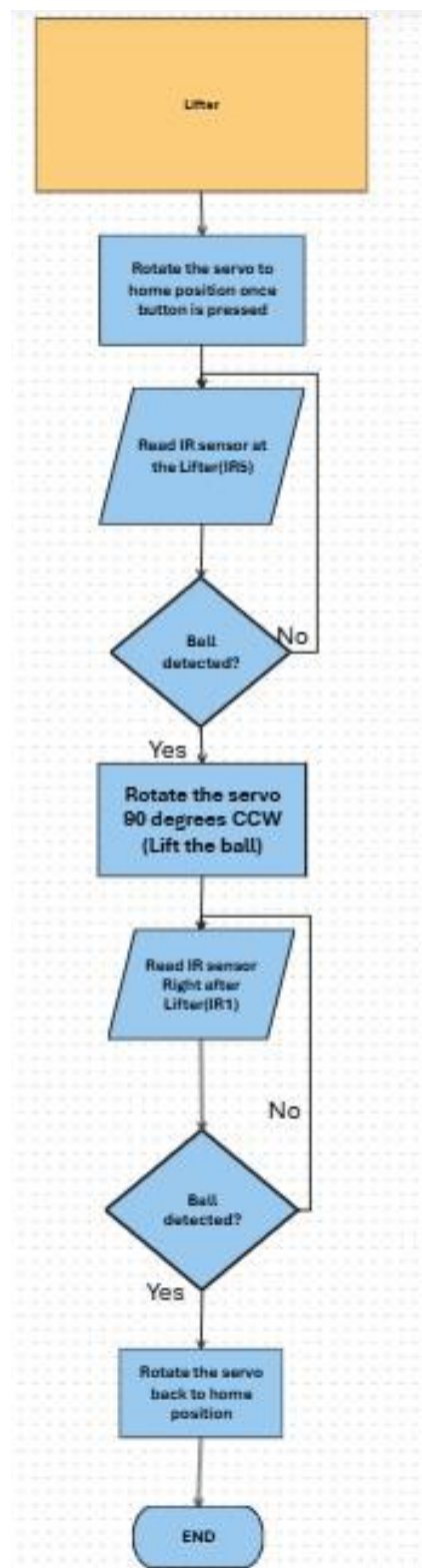
1. CAD Design



2. Wiring Diagram



3. Flowchart



4. Actuator Sizing

$$m_{\text{arm}} = 0.0306 \text{ kg}$$

Lifting

$$T_{\text{req}} = T_{\text{load}} + J_{\text{eq}} \cdot \alpha$$

$$m_{\text{box}} = 0.0162 \text{ kg}$$

$$m_{\text{ball}} = 0.0064 \text{ kg}$$

$$J_{\text{arm}} = \frac{1}{3} m l^2 = \frac{1}{2} (0.0306) \left(\frac{450}{1000} \right)^2 = 0.0031 \text{ kg} \cdot \text{m}^2$$

$$J_{\text{eq}} = (m_{\text{box}} + m_{\text{ball}}) l^2$$

$$= (0.0162 + 0.0064) \left(\frac{472.5}{1000} \right)^2 = 0.005 \text{ kg} \cdot \text{m}^2$$

$$T_f = F \times D$$

$$= 0.3 \left(\frac{181.5}{1000} \right) + (0.2177) \left(\frac{417.8}{1000} \right)$$

$$= 0.145 \text{ N} \cdot \text{m}$$

$$J_{\text{total}} = 0.0081 \text{ kg} \cdot \text{m}^2$$

$$\alpha = \frac{4\theta}{t^2} \quad \theta = \frac{\pi}{2} \quad \text{time} = 0.5$$

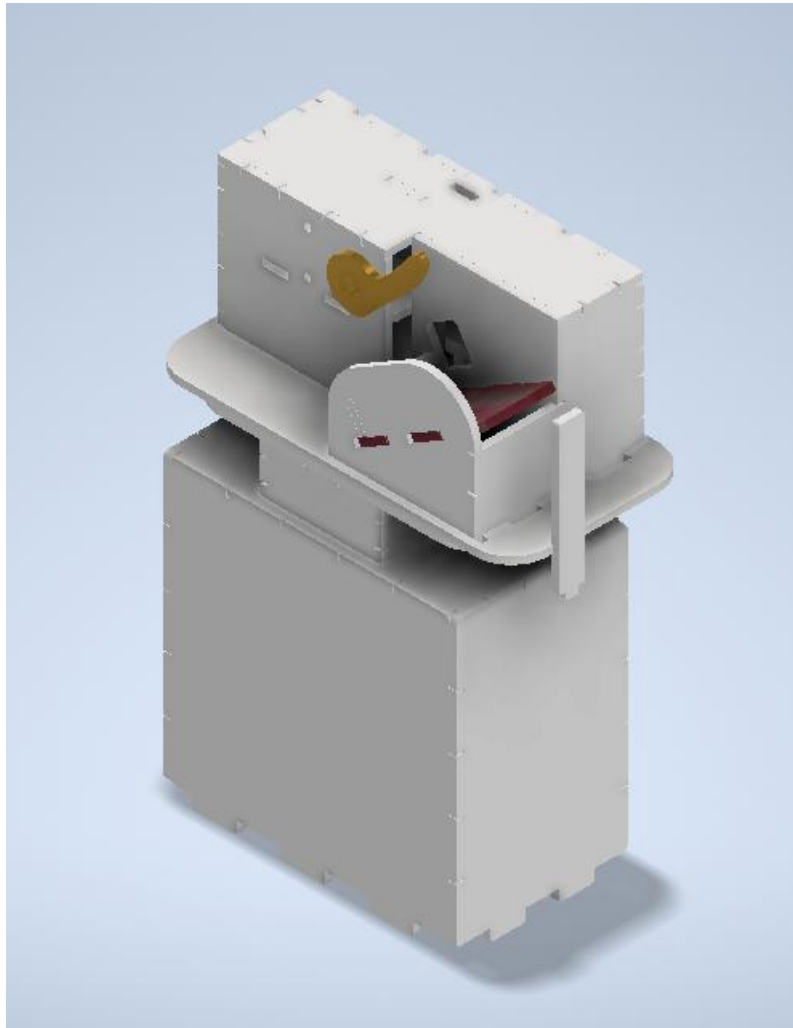
$$\frac{4(\frac{\pi}{2})}{(0.5)^2} = 8\pi$$

$$T_{\text{req}} = 0.145 + (0.0081)(8\pi) = 0.348 \text{ N} \cdot \text{m}$$

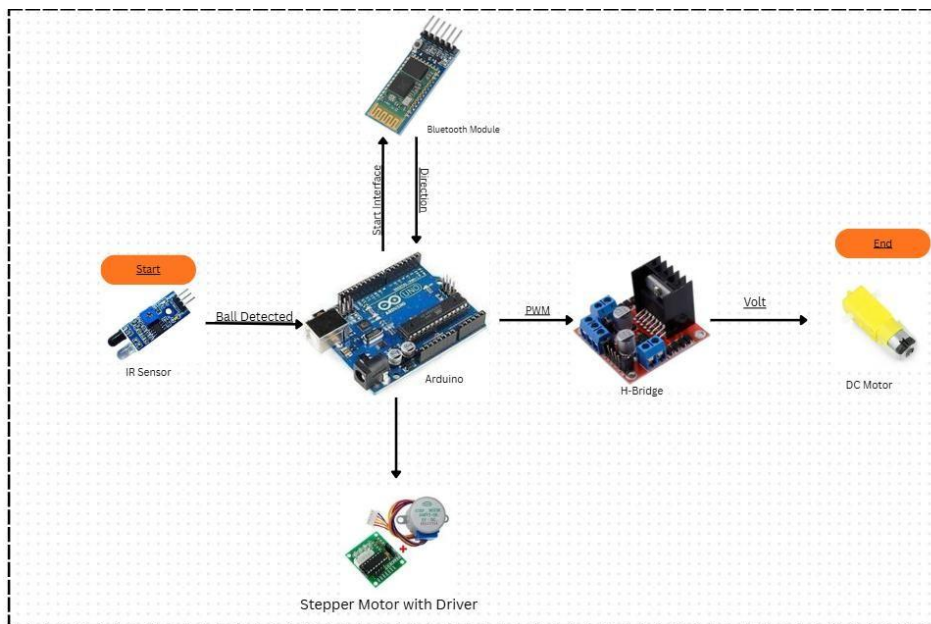
$$\text{Safety Factor} = 0.348 \times 1.5 = 0.522 \text{ N} \cdot \text{m}$$

Shooting mechanism

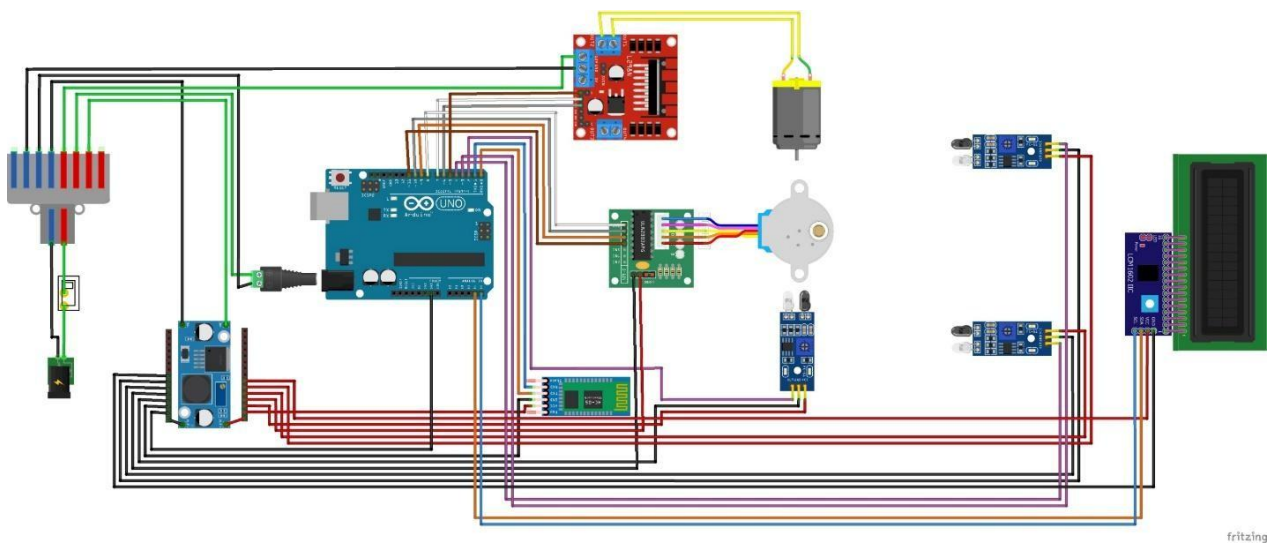
1. CAD Design



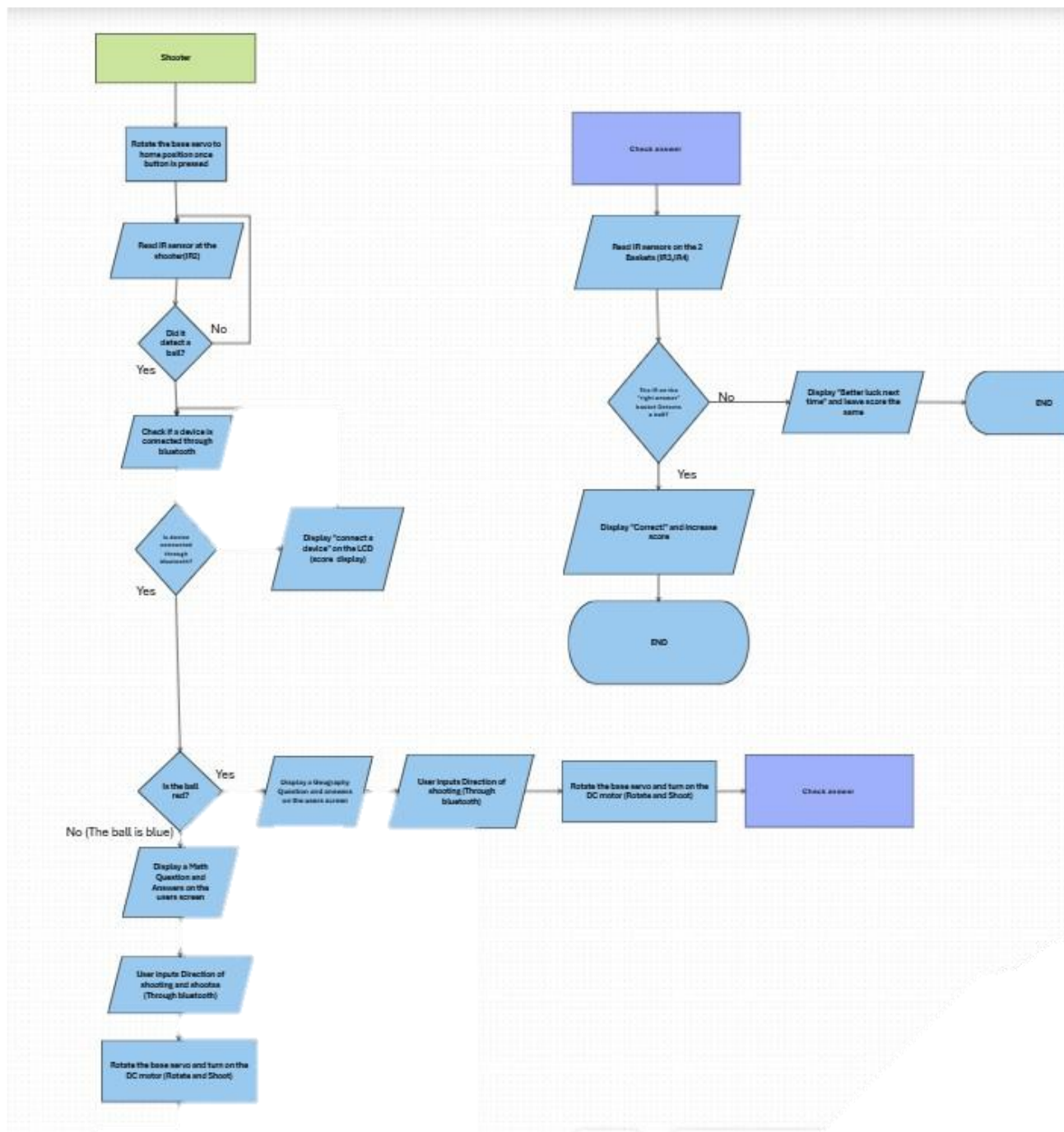
2. Block Diagram



3. Wiring Diagram



4. Flowchart



5. Projectile Calculations:

a) Problem Statement:

A DC motor with an arm shoots a 6 grams ball with the following conditions:

Projectile specs:

- Initial Height (y_0): 3.9 cm (0.039 m) (from the base of the motor)
- Horizontal Distance (x): 23 cm (0.23 m)
- Time of Flight (t_{total}): 0.27s
- Launch Angle (θ): 30°
- Ball mass (m_{ball}): 6g (0.006 kg)

kg) Arm specs:

- Length (L): 3.6 cm (0.036 m)
- Mass (m_{arm}): 1 g (0.001 kg)

b) Objectives:

- Determine initial velocity (V_x , V_y) of the ball.
- Calculate maximum height (H).
- Compute force applied by the arms (F_{avg}), and needed Torque (T_{avg}).
- Acceleration Components (A_x , A_y).

c) Step-by-Step Calculations:

1.1 Initial Velocity Components (V_x , V_y)

The ball moves horizontally 0.23 m in 0.27 s before reaching the datum (the base of the motor)

a) Horizontal Velocity (V_x):

$$X = V_x * t$$
$$V_x = \frac{X}{t} = \frac{0.23}{0.19} \approx 1.2 \text{ m/s}$$

b) Vertical Velocity (V_y):

$$y = y_0 + V_y * t - \frac{1}{2}(gt^2)$$

$$0 = 0.039 + V_y * 0.19 - (.5 * 9.81 * (0.19)^2)$$

$$V_y = ((0.5 * 9.81 * (0.19)^2) - 0.039) / 0.19 \approx 0.72 \text{ m/s}$$

c) Total Initial Velocity (V):

$$V = \sqrt{V_x^2 + V_y^2} = \sqrt{1.2^2 + 0.72^2} \approx 1.4 \text{ m/s}$$

d) Launch Angle (θ):

$$\theta = \tan^{-1}\left(\frac{V_y}{V_x}\right) = \tan^{-1}\left(\frac{0.72}{1.2}\right) \approx 30.9^\circ \text{ (Expected } 30^\circ)$$

1.2 Maximum Height (H):

a) Time to Reach H (t_{up}):

$$V_{y \text{ at } H} = V_y - g * t_{up}$$
$$t_{up} = \frac{V_y - 0}{g} = \frac{0.72}{9.81} \approx 0.073 \text{ s}$$

b) Calculating H:

$$H = y_0 + V_y * t_{up} - (0.5 * g * t_{up}^2)$$

$$H = 0.039 + 0.72 * 0.073 - (.5 * 9.81 * (0.073)^2) \approx 0.065 \text{ m} \approx 6.5 \text{ cm}$$

1.3 Torque Applied by Arm (Impulse)

a) Impulse (J):

$$J = \Delta P = m_{ball} * V = 0.006 * 1.4 = 8.4 \times 10^{-3} \text{ Kg.m/s}$$

b) Average Force (F_{avg}):

$$F_{avg} = \frac{J}{\Delta t} = \frac{.0084}{0.01} = 0.84 \text{ N}$$

c) Calculating Torque (T_{avg}):

$$T = F_{avg} * L = 0.84 * 0.036 = 0.03 \text{ N.m}$$

1.4 Acceleration Components (A_x, A_y):

a) Average Acceleration (A_{avg}):

$$F_{avg} = m_{ball} * A_{avg}$$

$$A_{avg} = \frac{F_{avg}}{m_{ball}} = \frac{.84}{0.006} \approx 140 \text{ m/s}^2$$

b) Acceleration Components (A_x, A_y):

$$A_x = A_{avg} \sin(\theta) = 140 * \cos(30.9) = 120.1 \text{ m/s}^2$$

$$A_y = A_{avg} \cos(\theta) = 140 * \sin(30.9) = 71.9 \text{ m/s}^2$$

d) Summary of Results:

Horizontal Velocity (V_x)	1.2 m/s
Vertical Velocity (V_y)	0.72m/s
Total Initial Velocity (V_0)	1.4 m/s
Launch Angle (θ)	30.9° (Expected 30)
Maximum Height (H)	6.5cm
Average Force (F_{avg})	0.84 N
Horizontal Acceleration A_x	120.1 m/s ²
Vertical Acceleration A_y	71.9 m/s ²

Figure 15, Summary of Results

6. Actuator Sizing:

a) Problem Statement:

- Using a PM DC motor, it needs to launch a 6 g ball for 23 cm, what is the specs of the needed DC motor.
- Using a Stepper motor, we need to rotate a 150 g base of the shooter for certain angles, what are the specs of the needed stepper motor

b) Objectives:

- Determine the needed torque for launching the ball.
- Select the DC motor.
- Determine the needed torque for rotating the base.
- Select the Stepper motor.

c) Step by Step Sizing:

1.1 Calculating needed torque for launching the ball (T_{req}):

$$T_{req} = (J * \theta^{\circ}) + T_{res}$$

a) Calculating Inertia Effective (J_{eff}):

$$J_{eff} = J_{load} + J_{motor}$$

$$\text{Let } J_{motor} = J_{load} \rightarrow J_{eff} = 2 * J_{load}$$

$$J_{load} = J_{origin} = J_{arm} + md^2 \text{ (Using Parallel Axis Theorem)}$$

Note: Knowing the Arm Inertia (J_{arm}) Using the software used to design the arm (Autodesk Inventor)

And the COG to determine the distance between the COG of the arm and the Origin

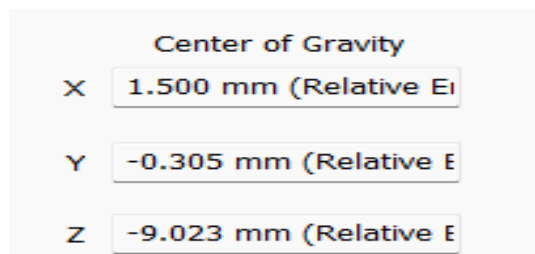


Figure 18, Shooter Arm COG

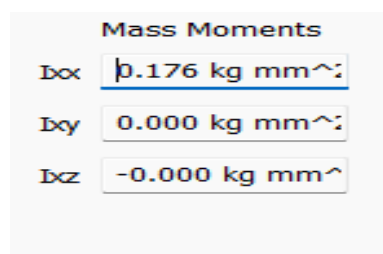
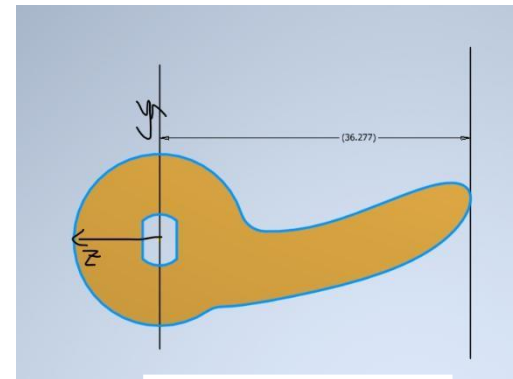


Figure 17, Shooter Arm Inertia



So, we got this $I_{xx} = 0.176 \text{ Kg.mm}^2 = 1.76 \times 10^{-5} \text{ kg.m}^2$,

$$d = \sqrt{y^2 + z^2} = \sqrt{(-0.305)^2 + (-9.023)^2} \approx 9.028 \text{ mm} = 0.009028 \text{ m}$$

$$J_{load} = J_{arm} + md^2 = 1.76 * 10^{-5} + 0.006 * (.009028)^2 \approx 1.8 * 10^{-5} \text{ Kg.m}^2$$

$$J_{eff} = 2 * J_{load} = 2 * 1.8 * 10^{-5} = 3.6 * 10^{-5} \text{ kg.m}^2$$

Form 3.1 section we got the initial velocity of 1.4 m/s

So we can calculate the angular velocity (ω_{launch}) form the equation:

$$\omega_{launch} = \frac{V^{\circ}}{r} = \frac{1.4}{0.036} = 38.88 \text{ rad/s}$$

Then we can build the motion profile according to ω_{launch}

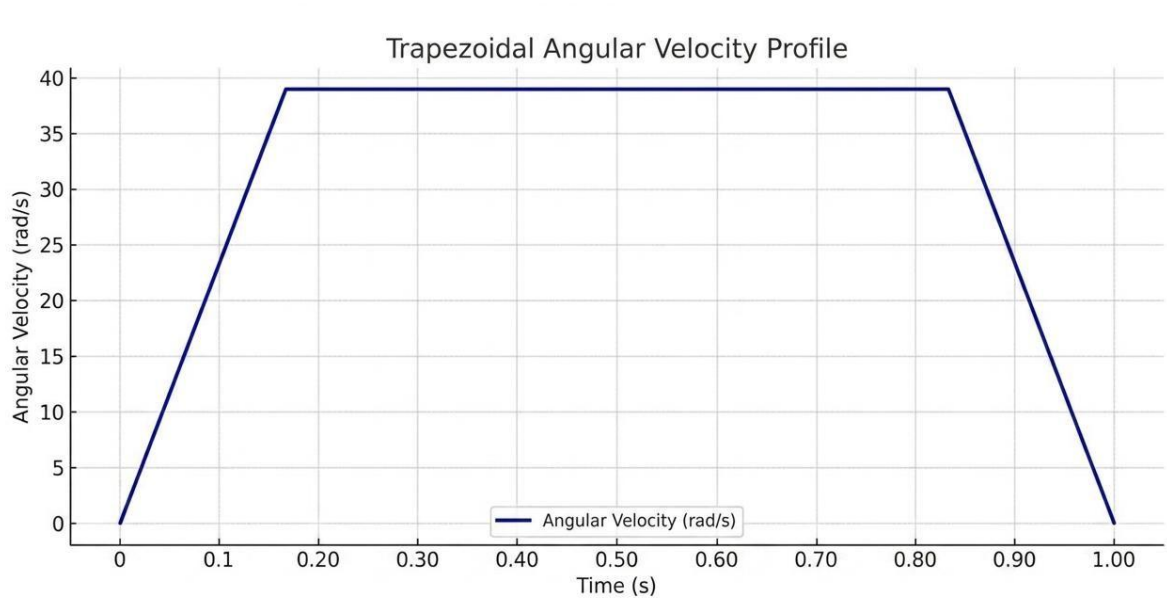


Figure 19, Trapezoidal Motion Profile of DC Motor

Where $t_{acc} = 0.15s$, $t_{constant} = 0.70s$, $t_{decc} = 0.15s$, and $t_{dwell} = 0.1s$.

Then we calculate the angular acceleration (θ°):

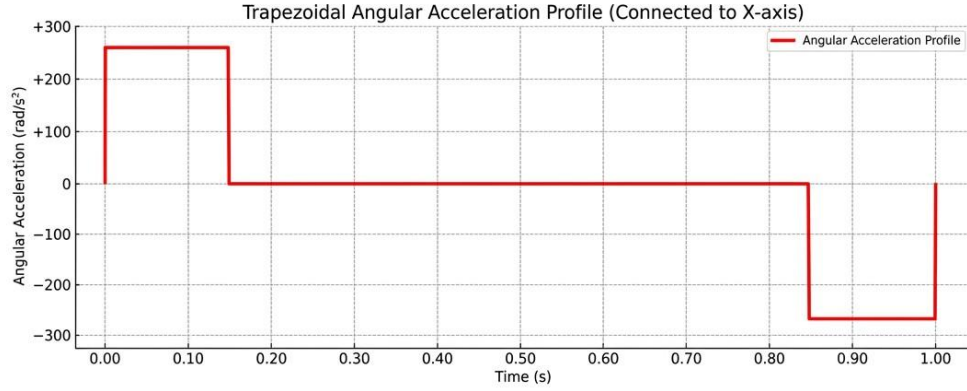


Figure 20, Acceleration Diagram of DC Motor.

$$\theta^{\circ}_{acc} = \alpha_{acc} \approx 259.253 \text{ rad/s}^2$$

$$\theta^{\circ}_{const} = \alpha_{const} = 0 \text{ rad/s}^2$$

$$\theta^{\circ}_{decc} = \alpha_{decc} \approx -259.253 \text{ rad/s}^2$$

Note: The T_{res} is neglectable, so we will consider it in the safety factor.

Now, We can calculate the Torque Required (T_{req}):

$$T_{req} = (J_{eff} * \theta^{\circ})$$

$$(T_{req})_{acc} = 3.6 * 10^{-5} * 259.253 \approx 0.0093 \text{ N.m} = T_{peak}$$

$$(T_{req})_{const} = 3.6 * 10^{-5} * 0 \approx 0 \text{ N.m}$$

$$(T_{req})_{decc} = 3.6 * 10^{-5} * -259.253 \approx -0.0093 \text{ N.m}$$

$$T_{rms}$$

$$= \sqrt{\frac{((T_{req})_{acc})^2 * t_{acc} + ((T_{req})_{const})^2 * t_{const} + ((T_{req})_{decc})^2 * t_{decc} + ((T_{req})_{dw})^2 * t_{dw}}{t_{acc} + t_{const} + t_{decc} + t_{dw}}}$$

$$= \sqrt{\frac{(0.0093^2 * 0.15) + (0^2 * 0.7) + (-0.0093^2 * 0.15) + (0^2 * 0.1)}{0.15 + 0.7 + 0.15 + 0.1}}$$

$$\approx 0.00486 \text{ N.m}$$

1.2 Select the DC motor:

We got that $(T_{req})_{peak} = 0.0093 \text{ N.m}$ & $T_{rms} = 0.00486 \text{ N.m}$

After multiplying at $n=20$ as a safety factor because we neglected the resistance torque (T_{res}) we need a DC motor with Torque of 0.186 N.m

We found that the M1-1-Q1-DC-Motor-4WD can be used on its maximum torque at 12V and it is equal 0.3 N.m

Knowing that the motor will not work at 100% efficiency so let assume $\eta = 0.6$

So the generated torque will be 0.18 N.m which is enough for our application.



Figure 21, M1-1-Q1-DC-Motor-4WD

1.3 Calculating needed torque for rotating the base (T_{req}):

$$T_{req} = (J * \theta^{\circ}) + T_{res}$$

a) Calculating Inertia Effective (J_{eff}):

$$J_{eff} = J_{load} + J_{motor}$$

Let $J_{motor} = J_{load} \rightarrow J_{eff} = 2 * J_{load}$

$$J_{load} = J_{origin} = J_{base} + md^2 \text{ (Using Parallel Axis Theorem)}$$

As shown in figure 22, the base has 14 items on it so it will be a long process to compute the inertia of each component so for simplifying we will represent the base and its components as a cylinder with the outer dimensions of the base which are $d = 75 \text{ mm}$, and $L = 150 \text{ mm}$

Knowing that the mass of the base with the components is 150 gm

So, we got this $I = 1.05 \times 10^{-4} \text{ kg.m}^2$,

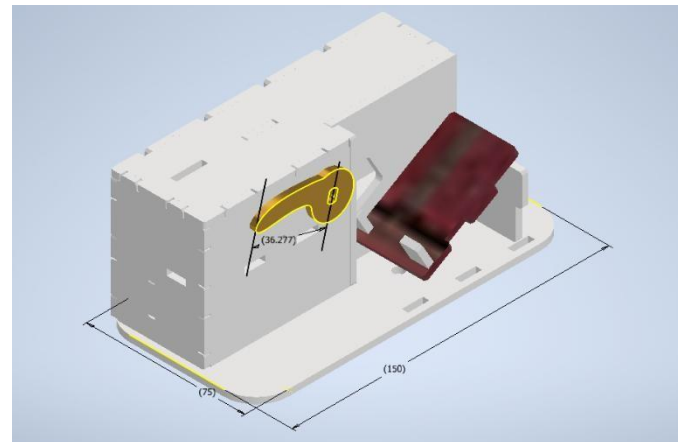


Figure 22, the base & all parts on it

The COG of the cylinder is be above the shaft of the motor with the radius of the cylinder so
 $d = 75/2 \text{ mm} = 0.0375 \text{ m}$

$$J_{load} = J_{base} + md^2 = 1.05 * 10^{-4} + 0.15 * (0.0375)^2 \approx 3.16 * 10^{-4} \text{ Kg.m}^2$$

$$J_{eff} = 2 * J_{load} = 2 * 3.16 * 10^{-4} = 6.3 * 10^{-4} \text{ kg.m}^2$$

We have set the speed of the stepper motor in the code to **10 rpm**

So we can calculate the angular velocity ($\omega_{rotating}$) form the equation:

$$\omega_{rotating} = \frac{2\pi * RPM}{60} = \frac{2\pi * 10}{60} = 1.047 \text{ rad/s}$$

Then we can build the motion profile according to $\omega_{rotating}$

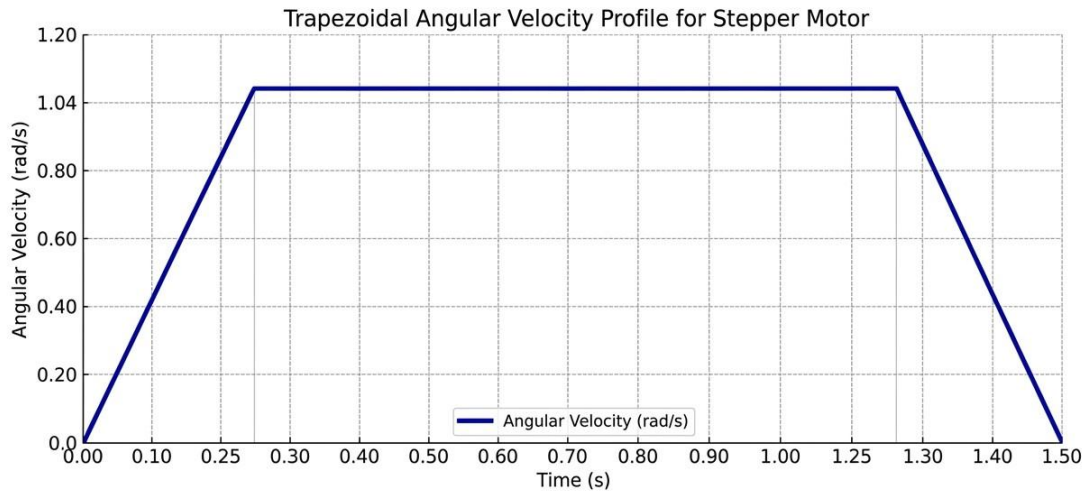


Figure 23, Trapezoidal Motion Profile of Stepper Motor

Where $t_{acc} = 0.25 \text{ s}$, $t_{constant} = 1.0 \text{ s}$, $t_{decc} = 0.25 \text{ s}$, and $t_{dwell} = 0.1 \text{ s}$

Then we calculate the angular acceleration (θ''):

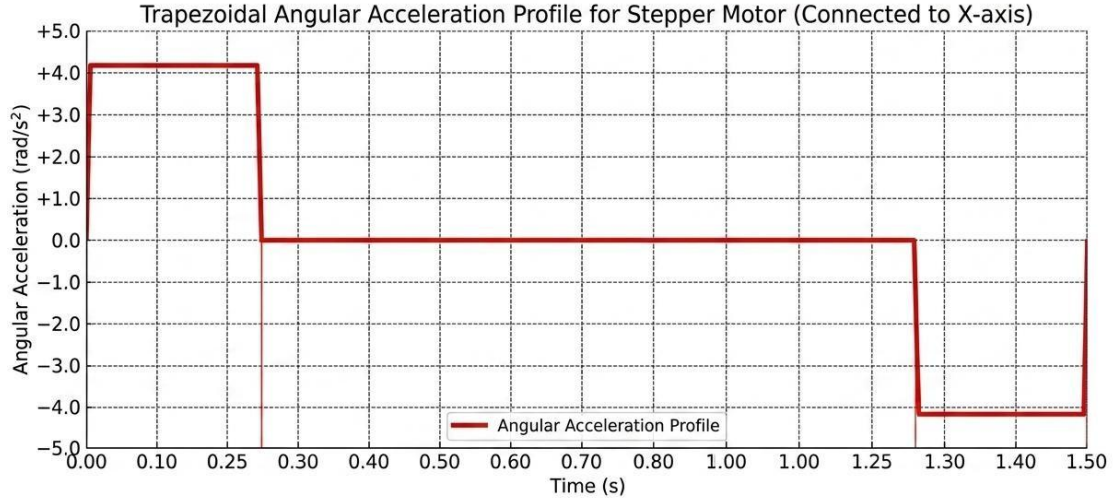


Figure 24, Acceleration Diagram of Stepper Motor.

$$\theta^{\circ}_{acc} = \alpha_{acc} \approx 4.16 \text{ rad/s}^2$$

$$\theta^{\circ}_{const} = \alpha_{const} = 0 \text{ rad/s}^2$$

$$\theta^{\circ}_{decc} = \alpha_{decc} \approx -4.16 \text{ rad/s}^2$$

Note: The T_{res} is neglectable, so we will consider it in the safety factor.

Now, We can calculate the Torque Required (T_{req}):

$$T_{req} = (J_{eff} * \theta^{\circ})$$

$$(T_{req})_{acc} = 6.3 * 10^{-4} * 4.16 \approx 0.0026 \text{ N.m} = T_{peak}$$

$$(T_{req})_{const} = 6.3 * 10^{-4} * 0 \approx 0 \text{ N.m}$$

$$(T_{req})_{decc} = 6.3 * 10^{-4} * -4.16 \approx -0.0026 \text{ N.m}$$

$$\begin{aligned}
 T_{rms} &= \sqrt{\frac{((T_{req})_{acc})^2 * t_{acc} + ((T_{req})_{const})^2 * t_{const} + ((T_{req})_{decc})^2 * t_{decc} + ((T_{req})_{dw})^2 * t_{dw}}{t_{acc} + t_{const} + t_{decc} + t_{dw}}} \\
 &= \sqrt{\frac{(0.0026^2 * 0.25) + (0^2 * 1.0) + (-0.0026^2 * 0.25) + (0^2 * 0.1)}{0.25 + 1.0 + 0.25 + 0.1}} \\
 &\approx 0.00145 \text{ N.m}
 \end{aligned}$$

1.4 State the Stepper motor:

We got that $(T_{req})_{peak} = 0.0026 \text{ N.m}$ & $T_{rms} = 00145 \text{ N.m}$

We found that the 28-BYJ-48 Stepper motor When it works on 12V it can Provide up to 0.0043 N.m Which is more than enough for our application and we can consider the excess in torque as safety factor $n = 2$.



Conclusion

The successful completion of this major task served as a comprehensive application of mechatronics engineering fundamentals, bridging the gap between theoretical design and physical implementation. By integrating mechanical design, electronics, and embedded programming, we developed a functional automated system capable of precise ball detection and launching.

Throughout the project, several key milestones were achieved:

- **Mechanical Synthesis:** Using Autodesk Inventor, we designed and simulated a snap-motion rack-and-pinion mechanism, ensuring that the mechanical components could withstand the required stresses while maintaining smooth motion.
- **Hardware Integration:** The system utilized an Arduino-based control unit paired with IR sensors for real-time feedback, demonstrating the importance of sensor accuracy in autonomous systems.
- **Fabrication Excellence:** The transition from CAD to reality involved 3D printing and laser cutting, where we learned to account for material tolerances and the effective use of mechanical fasteners like M3 screws and heat-set inserts.

Ultimately, this project reinforced the necessity of a multidisciplinary approach. It provided invaluable experience in troubleshooting complex electromechanical interfaces and refined our ability to manage a project from the initial concept phase through to a fully realized, integrated prototype.